



NATURAL LANGUAGE PROCESSING

Project Title

MedTriage: Symptom-to-Department Triage Using TF-IDF, Logistic Regression and Calibrated Confidence

Programme	B.Sc. (Artificial Intelligence)
Academic Year	2026-27
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1. Abstract

Hospitals routinely send patients to the wrong first desk because the first contact is based on a short free-text complaint. This project builds **MedTriage**, an NLP classifier that reads a free-text clinical description and suggests one of 11 hospital departments. I used the public MTSamples medical-transcription dataset (4,999 records), filtered it to 1,883 records belonging to 11 department-type specialties, and trained a TF-IDF (unigram+bigram) representation with a class-balanced Logistic Regression. On a stratified held-out set of 377 records the model reaches **79.8% accuracy** and **0.76 macro-F1**. A subgroup audit shows that accuracy drops from 83.8% on long descriptions to 76.0% on short ones, and that Pediatrics is the weakest department (F1 0.39). An explainability check of the learned coefficients shows that most departments are driven by sensible medical vocabulary. During testing of the Gradio demo I found that the raw model was strongly *under-confident* (mean confidence 0.37, Expected Calibration Error 0.426) even when it was correct. Temperature scaling ($T = 2.95$), learned only on the training data, reduced the ECE to 0.079 without changing a single prediction, and a low-confidence fallback at 0.50 routes uncertain cases to human triage. The system is a classroom prototype and not a clinical tool.

2. Introduction

Patient routing is the first decision in a hospital visit. A receptionist or call-centre agent reads a short description such as "chest pain radiating to the left arm" and decides between cardiology, orthopedics, neurology and so on. Wrong routing wastes time and delays care. Because the input is unstructured text, this is a natural text-classification problem.

I chose this problem because it is easy to explain, has a clear label for every example, and exposes the questions this course cares about: how does a model behave on short versus long text, on rare versus common classes, and how much should we trust its confidence? NLP provides the pipeline that turns raw text into numerical features (TF-IDF) and a classifier that maps those features to a department.

3. Problem Statement and Objectives

Problem statement. Given a free-text clinical or symptom description in English, predict the most appropriate hospital department out of 11 candidates and report how confident the model is.

Objectives

- Build an end-to-end NLP pipeline (data cleaning, TF-IDF, classifier, evaluation) on a real dataset.
- Evaluate beyond overall accuracy: per-department metrics and at least two text subgroups (bias and performance audit).
- Add a simple explainability analysis using the model's top features.
- Make the confidence scores trustworthy (calibration) and handle uncertain cases safely.
- Deliver a working live demo, a model card, and an honest statement of limitations.

4. Dataset Description

Source: MTSamples (Medical Transcriptions), public sample transcriptions from mtsamples.com, distributed on Kaggle/GitHub as *mtsamples.csv*. The records are de-identified sample reports, not

real patient files. **Raw size:** 4,999 records and 6 columns (index, description, medical_specialty, sample_name, transcription, keywords). **Language:** English.

Label: the *medical_specialty* column, mapped to 11 hospital departments. **Input used:** the *description* field, a one- or two-sentence summary of the case (median 15 words in the test set). The raw data has 40 specialties, many of which are not departments (for example "Discharge Summary", "Letters", "Office Notes"); these were excluded. Class sizes are strongly imbalanced, as Table 1 and Figure 1 show.

Department	Records
Cardiology	372
Orthopedics	355
Gastroenterology	230
Neurology	223
Gynecology	160
Urology	158
ENT	98
Ophthalmology	83
Nephrology	81
Pediatrics	70
Psychiatry	53
Total	1,883

Table 1: Records per department after filtering.

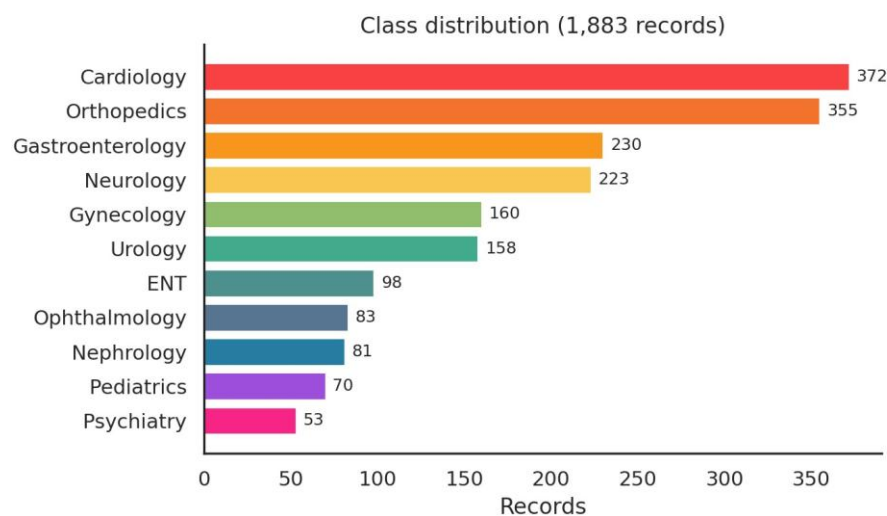


Figure 1: Class distribution. Cardiology has seven times more records than Psychiatry.

A second noteworthy property is repetition: only 1,745 of the 1,883 descriptions are unique, and 44 of the 377 test descriptions also occur in the training set (often generic text that appears under different specialties). This matters for the error analysis in Section 11.

5. Data Preprocessing

- **Department filtering and mapping.** Only specialties that correspond to a real hospital department were kept and renamed (for example "Cardiovascular / Pulmonary" to Cardiology, "Orthopedic" to Orthopedics, "Obstetrics / Gynecology" to Gynecology). Each kept class has at least 40 records.
- **Missing text.** Rows with a missing description were dropped; whitespace around labels was stripped.
- **Tokenisation and normalisation.** Handled inside the TF-IDF vectoriser: lower-casing, word tokenisation, punctuation removal, English stop-word removal. No stemming was used, so that clinical terms such as "cholecystectomy" stay intact.
- **Train/test split.** 80/20, stratified by department, random_state = 42, giving 1506 training and 377 test records.
- **Input choice (important).** My first version trained on the description *plus* the dataset's keyword column (with department-name words removed to avoid label leakage). Since a user of the app never types keywords, the final version uses the description only, so training and live input match (see Section 10.6).

6. Feature Representation

Text is represented with **TF-IDF** (term frequency - inverse document frequency) over unigrams and bigrams, limited to the 8,000 most frequent terms, with min_df = 2 and sublinear term frequency ($1 + \log \text{tf}$). TF-IDF was chosen because (i) the descriptions are short and keyword-driven, so word presence is a strong signal; (ii) bigrams capture phrases such as "otitis media" or "chest pain"; (iii) the resulting linear model is fast and directly interpretable through its coefficients, which the audit requirement needs. Its weakness is that it has no notion of meaning: "cardiac" and "heart" are unrelated features unless both appear in training.

7. Proposed Methodology / System Architecture

The workflow has an offline training part and a live inference part (Figure 2). The key addition over a plain classifier is the calibration step, which converts raw scores into trustworthy probabilities, and the fallback rule that flags uncertain predictions.

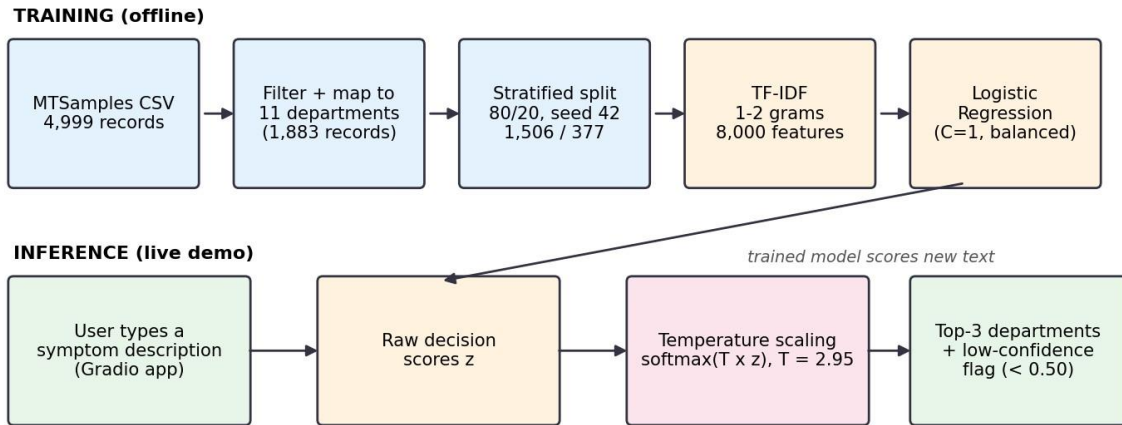


Figure 2: End-to-end architecture of MedTriage.

8. NLP Model / Technique Used

Classifier. Multinomial Logistic Regression on the TF-IDF vectors. Each department k gets a weight vector w_k ; the decision score is $z = w_k \cdot x + b_k$ and the probability is $\text{softmax}(z)$. Class k weighting ('balanced') makes errors on small departments cost more, so Psychiatry or Pediatrics are not ignored.

Component	Setting
Vectoriser	TfidfVectorizer: ngram_range (1,2), max_features 8000, min_df 2, sublinear_tf True, stop_words 'english'
Classifier	LogisticRegression: C = 1, class_weight 'balanced', max_iter 3000, L2 penalty (default)
Calibration	Temperature scaling, T = 2.95 (fitted on 5-fold out-of-fold training scores)
Fallback	If top probability < 0.50, flag for human triage
Seed	random_state = 42 (split and cross-validation)

Table 2: Model and hyper-parameter settings.

Temperature scaling. Plain Logistic Regression with L2 regularisation ($C = 1$) on sparse TF-IDF vectors gives small score magnitudes, so the softmax is flat. Temperature scaling (Guo et al., 2017) multiplies the scores by a single number T before the softmax: $p = \exp(T \cdot z_j) / \sum_k \exp(T \cdot z_k)$. Since $T > 0$ does not change which class has the largest score, accuracy is untouched; only the confidence changes. T was found by minimising log-loss on out-of-fold decision scores from the training set, so the test set stayed unseen.

9. Implementation

The system is implemented in Python 3 in a Google Colab notebook using pandas and numpy (data), scikit-learn (TF-IDF, Logistic Regression, metrics, cross-validation), scipy (temperature search), matplotlib and seaborn (figures) and Gradio (web interface). The notebook is organised in numbered

stages: setup, loading, exploration, cleaning and mapping, split, model, calibration, evaluation, audit, explainability and demo.

The Gradio app has a free-text box, an *Analyze* button, quick test examples and a result panel showing the top-3 departments with confidence bars, inference time, and a warning banner when confidence is below 0.50. Figure 3 shows the first version of the interface, captured before calibration was added: the correct department is ranked first but the displayed confidence is only 23.7%. This observation triggered the calibration work described in Section 10.5.

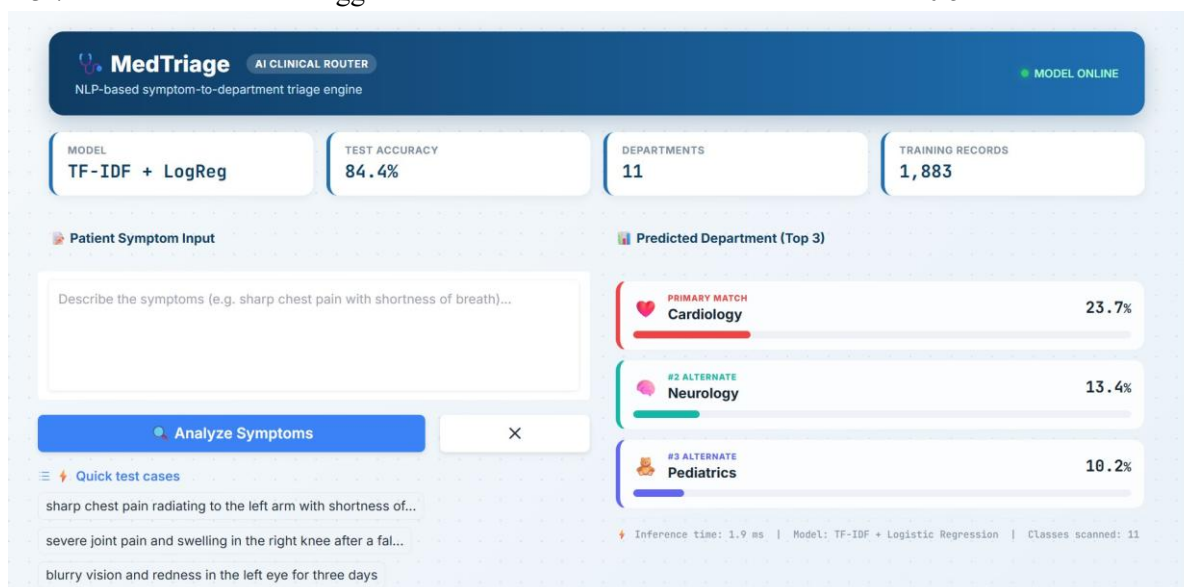


Figure 3: Demo before calibration (top class shown at only 23.7%). A post-fix screenshot is given in the Annexure.

10. Results and Evaluation

10.1 Overall performance

On the 377 held-out records the final model reaches **accuracy 79.8%**, **macro-F1 0.76** and weighted-F1 0.80. Table 3 gives the per-department breakdown and Figures 4-5 show the confusion matrix and F1 scores.

Department	Precision	Recall	F1	Support
Cardiology	0.90	0.86	0.88	74
ENT	0.79	0.75	0.77	20
Gastroenterology	0.86	0.83	0.84	46
Gynecology	0.78	0.88	0.82	32
Nephrology	0.81	0.81	0.81	16
Neurology	0.65	0.67	0.66	45
Ophthalmology	0.76	0.76	0.76	17
Orthopedics	0.87	0.86	0.87	71
Pediatrics	0.35	0.43	0.39	14

Psychiatry	0.60	0.90	0.72	10
Urology	0.92	0.75	0.83	32
Macro avg	0.76	0.77	0.76	377
Weighted avg	0.81	0.80	0.80	377

Table 3: Per-department results on the test set.

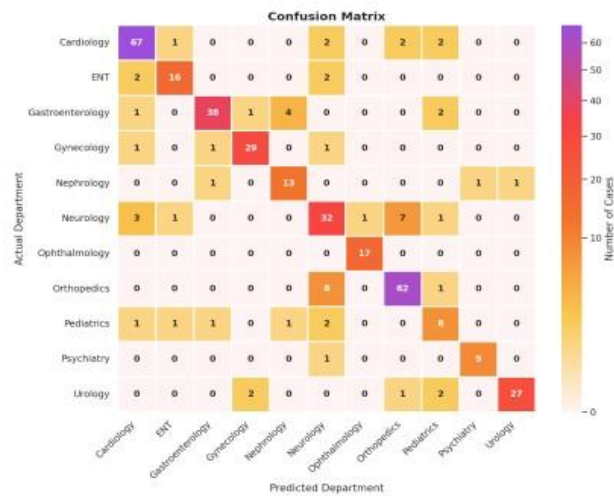


Figure 4: Confusion matrix on the test set.

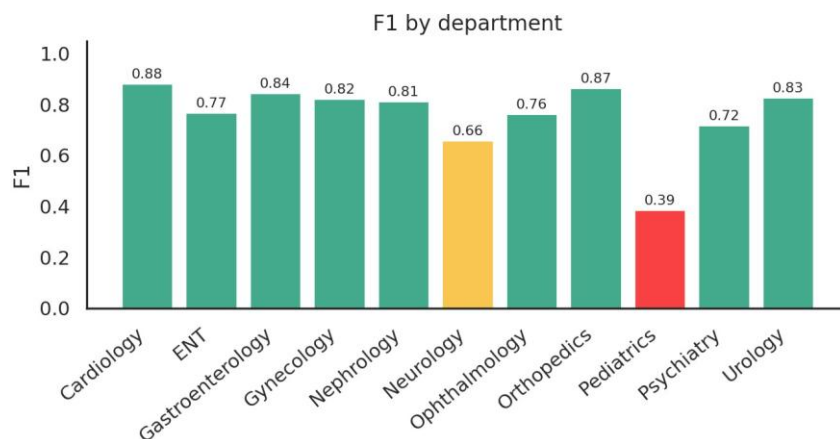


Figure 5: F1 score per department.

Orthopedics, Cardiology and Gastroenterology are strong (large classes, distinctive vocabulary). Pediatrics (F1 0.39) and Neurology (F1 0.66) are weak: Pediatrics has only 56 training records and

its descriptions use general words, and Neurology shares vocabulary (spine, MRI, headache) with Orthopedics, Ophthalmology and others.

10.2 Bias and performance audit across text subgroups

Subgroup	Definition	n	Accuracy
Short (≤ 15 words)	Word count at or below test median (15)	192	76.0%
Long (> 15 words)	Word count above the median	185	83.8%
Narrative case style	Contains patient-case cues (year-old, presents, history of, he/she, patient...)	104	73.1%
Terse title/procedure style	Procedure/diagnosis titles without such cues	273	82.4%

Table 4: Accuracy across text subgroups.

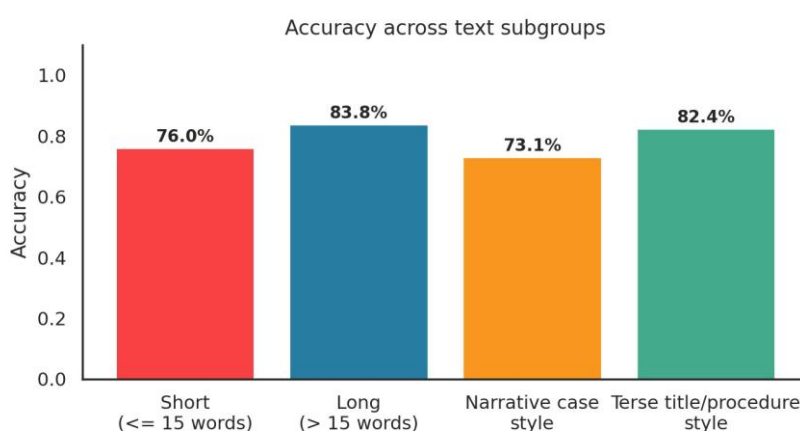


Figure 6: Accuracy across the four subgroups.

Key audit finding. Short descriptions are classified 7.7 points less accurately than long ones, and narrative patient-case text is 9.3 points less accurate than terse procedure-style titles. Narrative text contains many generic words ("the patient is a 45-year-old male who presents with...") that carry little department information, while procedure titles contain strongly specific terms ("cholecystectomy", "cataract"). The practical consequence is that users who describe symptoms in everyday narrative style get less reliable routing than users who use clinical terms.

10.3 Explainability: top features per department

For each department I listed the terms with the highest Logistic Regression coefficients (Table 5).

Department	Top terms (highest weight first)
Cardiology	chest, coronary, pulmonary, heart, bronchoscopy, lung, atrial, cardiac
ENT	nasal, ear, adenoidectomy, tonsillectomy, bilateral, otitis, otitis media, media
Gastroenterology	colonoscopy, esophagogastroduodenoscopy, dysphagia, cholecystectomy, laparoscopic, laparoscopic cholecystectomy, bowel, abdominal
Gynecology	breast, pregnancy, pelvic, hysterectomy, tubal, vaginal, ultrasound, delivery
Nephrology	renal, fistula, renal failure, nephrectomy, kidney, failure, pelvis, stent

Neurology	brain, mri, spine, craniotomy, eeg, cerebral, temporal, ct
Ophthalmology	eye, blepharoplasty, cataract, vitrectomy, right eye, lid, vision, sample
Orthopedics	knee, pain, fracture, right, foot, left, release, shoulder
Pediatrics	child, old, physical, month, child check, check, newborn, congestion
Department	Top terms (highest weight first)
Psychiatry	psychiatric, disorder, testing, depression, patient, evaluation, psychiatric consultation, admission
Urology	bladder, prostate, circumcision, inguinal, penile, cystoscopy, vasectomy, hematuria

Table 5: Highest-weight terms per department.

Most lists are medically sensible: Cardiology (chest, coronary, cardiac), ENT (ear, otitis, tonsillectomy), Ophthalmology (eye, cataract, vitrectomy), Urology (bladder, prostate, vasectomy). Two warnings appear. Pediatrics is driven by generic terms (child, old, check, month), explaining its weak F1. Ophthalmology contains the term 'sample', and Orthopedics has 'left' and 'right'; these are dataset artefacts (template wording) rather than medical signal, a small example of the model learning shortcuts.

10.4 Confidence before and after calibration

When I typed a description taken directly from the dataset into the demo, the correct department was returned but with a confidence of only about 20-40%. This is not a bug in the app; it is a property of the model. Table 6 quantifies it on the test set.

Measure	Raw model	Calibrated (T = 2.95)
Mean confidence (all predictions)	0.37	0.78
Mean confidence when correct	0.40	0.83
Mean confidence when wrong	0.27	0.60
Expected Calibration Error (lower is better)	0.426	0.079
Log-loss (lower is better)	1.23	0.72
Accuracy	79.8%	79.8% (identical)

Table 6: Effect of temperature scaling on the test set.

The raw model's average confidence (0.37) is far below its true accuracy (79.8%), so a prediction displayed as 25% is in reality right most of the time. After scaling, a displayed confidence is close to the real hit rate.

10.5 Why the confidence was low and how it was fixed

- **Cause 1: regularised linear model on sparse TF-IDF.** Each TF-IDF vector is L2-normalised and has only a few non-zero entries (a description has about 15 words), and $C = 1$ shrinks the weights. The class scores z therefore have a small spread, so the softmax output is flat across 11 classes. The ranking is right, the scale is wrong.

- **Cause 2: train/serve mismatch.** The first model was trained on description plus keywords but the demo supplied descriptions only, so most of the evidence the model expected was missing. The final model is trained on the same kind of input it receives.
- **Fix.** (a) train on description only; (b) temperature scaling fitted on out-of-fold training scores; (c) a low-confidence fallback. Simply increasing C was tested and rejected: C = 10 raised confidence but lowered accuracy (e.g. 79.8% to about 77%).

10.6 Fallback threshold and ablation

Confidence threshold	Cases auto-routed	Accuracy on those cases
≥ 0.40	89%	83.9%
≥ 0.50	83%	86.0%
≥ 0.60	77%	87.5%
≥ 0.70	69%	88.1%

Table 7: Coverage versus accuracy for the fallback rule (calibrated confidence). I use 0.50.

Model variant	Test input	Accuracy
v1: description + keywords	description + keywords	84.4%
v1: description + keywords	description only (what the demo gives)	80.1%
v2 (final): description only	description only	79.8%

Table 8: Ablation. The 84.4% figure was measured on input that the live demo never receives, so the honest figure for the deployed system is 79.8%.

Sample test-case outputs before and after calibration are listed in Annexure B.

11. Error Analysis / NLP Failure Analysis

The metric alone hides why the model fails. Table 9 lists real test-set errors with high confidence. In several of them the model's answer is arguably reasonable and the dataset label is the questionable part.

Description (test set)	True label	Predicted	Conf.
Cystourethroscopy, bilateral retrograde pyelogram, and transurethral resection of bladder tumor of 1.5 cm in size. Recurrent bladder tumor and history of bladder carcino	Nephrology	Urology	99%
The patient is a 4-month-old who presented with supraventricular tachycardia and persistent cyanosis.	Cardiology	Pediatrics	99%
Psychosocial evaluation of kidney donor. Questions - Answers	Nephrology	Psychiatry	98%
Direct laryngoscopy, rigid bronchoscopy and dilation of subglottic upper tracheal stenosis.	Cardiology	ENT	95%
Description (test set)	True label	Predicted	Conf.

CT abdomen and pelvis without contrast, stone protocol, reconstruction.	Gastroenterology	Nephrology	99%
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Table 9: Representative high-confidence errors.

- **Overlapping specialties.** "Transurethral resection of bladder tumor" is a urological procedure but is labelled Nephrology; "CT abdomen and pelvis, stone protocol" is labelled Gastroenterology but the vocabulary (stone, pelvis) is nephrology/urology. The model follows the words, the label follows the reporting physician.
- **Label definition.** The source specialty "Cardiovascular / Pulmonary" was mapped to Cardiology, so a bronchoscopy case is labelled Cardiology while the model predicts ENT.
- **Context the model cannot see.** "A 4-month-old with supraventricular tachycardia" is a cardiology case in a baby. The pediatric cue (age) and the cardiac cue (tachycardia) compete, and a bag-of-words model has no way to decide which is more important.
- **Short, imaging-only text.** Descriptions that only name a scan ("MRI C-spine") say where the scan is, not which department owns the patient.
- **Duplicated generic descriptions.** Among the 44 test descriptions that also appear in training, accuracy is only 43.2%, against 84.7% on the other 333, because identical text appears under different specialties. No model can resolve this; it is a ceiling set by the labels.

A further finding is that wrong predictions can still carry high confidence (mean 0.60 for wrong versus 0.83 for right predictions). Calibration fixes the average level, not individual overconfident mistakes; this is why the fallback and human oversight are necessary.

12. Ethical Considerations and Limitations

- **Not for clinical use.** This is a classroom prototype. A wrong department suggestion in a real hospital could delay care; any real use would need clinician oversight, a representative dataset and formal safety testing.
- **Dataset limitations.** MTSamples is a small collection of sample transcriptions, not a real patient population. It is imbalanced (Cardiology 372 vs Psychiatry 53), and labels are specialty tags rather than expert triage decisions.
- **Performance bias.** The audit shows lower accuracy for short and narrative text and for rare departments. People who describe symptoms in plain, non-clinical language or who need pediatric or mental-health care are served worse by this model.
- **Model limitations.** Pure word-matching: no negation ("no chest pain"), no synonyms, no understanding of severity or emergencies, English only, no spelling correction. It picks exactly one department.
- **Privacy.** The data are public, de-identified samples. A deployed version would handle sensitive personal health information and would need consent, secure storage and legal compliance.
- **Confidence.** Calibration makes the average confidence honest but individual predictions can still be confidently wrong.

13. Conclusion

I built a complete NLP pipeline that routes free-text clinical descriptions to 11 hospital departments with 79.8% accuracy and 0.76 macro-F1 on unseen data. The audit showed where the model is weak (short and narrative text, Pediatrics and Neurology) and the explainability check showed that it

mostly relies on meaningful medical vocabulary, with a few dataset shortcuts. The most useful lesson came from the demo: a model can be accurate and still report misleading confidence. Diagnosing the low confidence led to two corrections, matching the training input to the live input and applying temperature scaling, which cut calibration error from 0.426 to 0.079 without changing accuracy. The objectives were met; the main open problem is the ceiling imposed by overlapping specialties and noisy labels.

14. Future Scope

- Replace TF-IDF with contextual embeddings (for example a clinical BERT variant) and repeat the same subgroup audit for comparison.
- Collect more Pediatrics, Psychiatry and Nephrology examples, or use data augmentation, to reduce class imbalance.
- Allow multi-label output so that cases spanning two departments are routed to both.
- Add negation handling and spelling normalisation, and support Hindi/Marathi or code-mixed input.
- Evaluate with clinicians and with real-world triage notes before any practical use.

15. References

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- [6] MTSamples. Medical Transcription Samples. <https://www.mtsamples.com> (dataset alsodistributed as mtsamples.csv on Kaggle).
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- [8] McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*.

16. GitHub Repository

GitHub Repository Link: <https://github.com/lambdadesrushti/medtriage-nlp.git>

The repository contains the notebook (MedTriage_NLP.ipynb), the dataset file, the model card and this report, so the results can be reproduced by running all cells.

17. Team Contribution

Team Member	PRN	Specific Contribution
Srushti Lamnbdade	25030421021	Individual project: problem selection, data cleaning and mapping, model building, calibration fix, bias and explainability audit, Gradio demo, model card, report and presentation.

18. Annexure

Annexure A : Demo test cases before and after calibration

Input typed into the demo	Predicted	Raw conf.	Calibrated conf.
sharp chest pain radiating to the left arm with shortness of breath	Cardiology	39%	93%
severe joint pain and swelling in the right knee after a fall	Orthopedics	50%	99%
blurry vision and redness in the left eye for three days	Ophthalmology	67%	100%
sinus congestion and nasal blockage with facial pressure	ENT	50%	98%
ear pain and hearing loss for the past week	ENT	26%	70%
abdominal pain and nausea after eating	Gastroenterology	37%	94%

Table B1: The six built-in quick test cases. Predictions are identical before and after; only the confidence changes.

Descriptions sampled from the training data (the situation that originally looked wrong):

Description from training set	True	Predicted	Raw	Calib.
Spontaneous vaginal delivery. Term pregnancy at 40 and 3/7th weeks. On evaluation of triage, she was noted to be cont	Gynecology	Gynecology	57%	99%
Left distal both-bone forearm fracture. Closed reduction with splint application with use of image intensifier.	Orthopedics	Orthopedics	44%	97%
Primary low-transverse cesarean section.	Gynecology	Gynecology	66%	100%
He awoke one morning and had double vision. He states when he closed each eye, the double vision dissipated. The doubl	Ophthalmology	Ophthalmology	49%	99%
Evaluate for retroperitoneal hematoma, the patient has been following, is currently on Coumadin. CT abdomen without con	Gastroenterology	Nephrology	49%	96%
Successful stenting of the left anterior descending. Angina pectoris, tight lesion in left anterior descending.	Cardiology	Cardiology	28%	79%

Table B2: Random training-set descriptions.

Annexure B: Reproducibility checklist

- Dataset: mtsamples.csv (4,999 rows) from the repository link in the notebook.
- Filter: 11 mapped specialties, 1,883 rows; input column: description.
- Split: train_test_split(test_size=0.2, random_state=42, stratify=department).
- Model: TfidfVectorizer(max_features=8000, ngram_range=(1,2), stop_words='english', sublinear_tf=True, min_df=2) + LogisticRegression(C=1, class_weight='balanced', max_iter=3000).
- Calibration: T fitted on 5-fold out-of-fold decision scores from the training set only (StratifiedKFold, random_state=42), T = 2.95.
- Fallback threshold: 0.50 on calibrated top-1 probability.

Annexure C: Post-fix demo screenshot

